

Short Read Sequencing Workshop Day 2

Linux

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Turn and Talk to Your Neighbor: What do the following commands do?

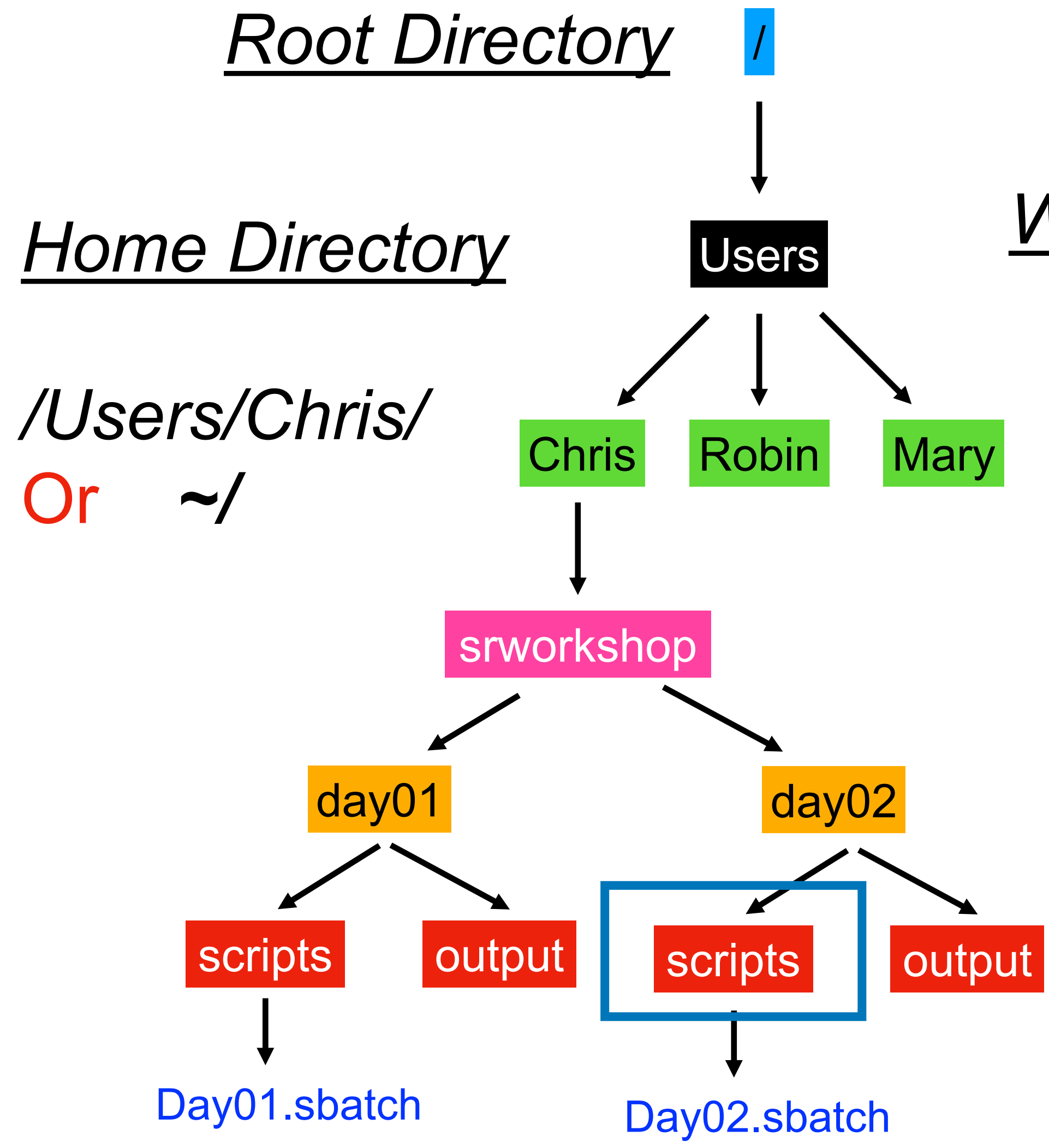
pwd

cd

ls

Navigating the Hierarchical Directory Structure

Linux Operating System File Structure



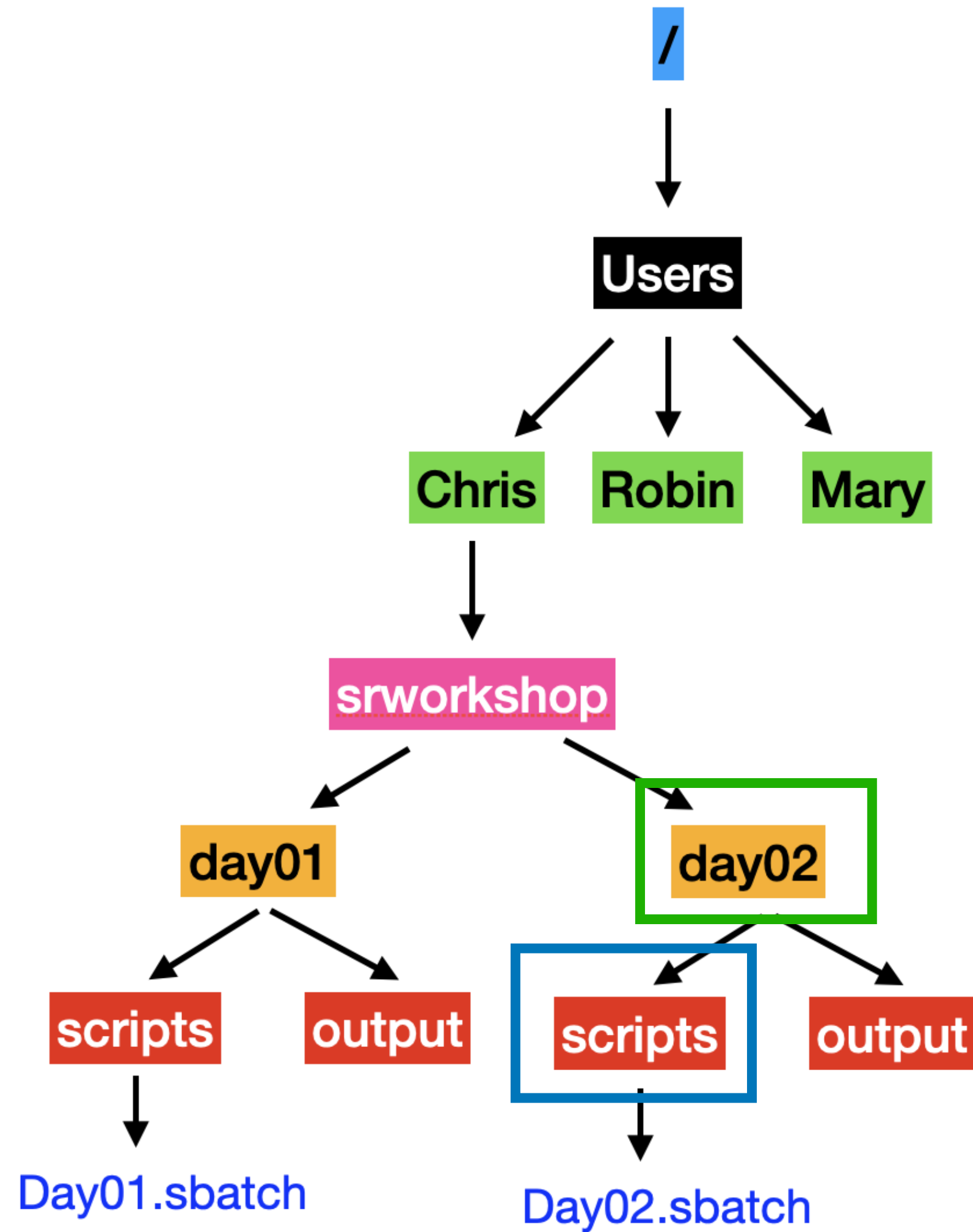
Working Directory: Current Directory you are in

/Users/Chris/srworkshop/day02/scripts

or *./*

Symbol for current working directory

Navigating the Hierarchical Directory Structure



Objective: Change from current working directory of **scripts** to **day02**

1. Absolute Path - path from root directory

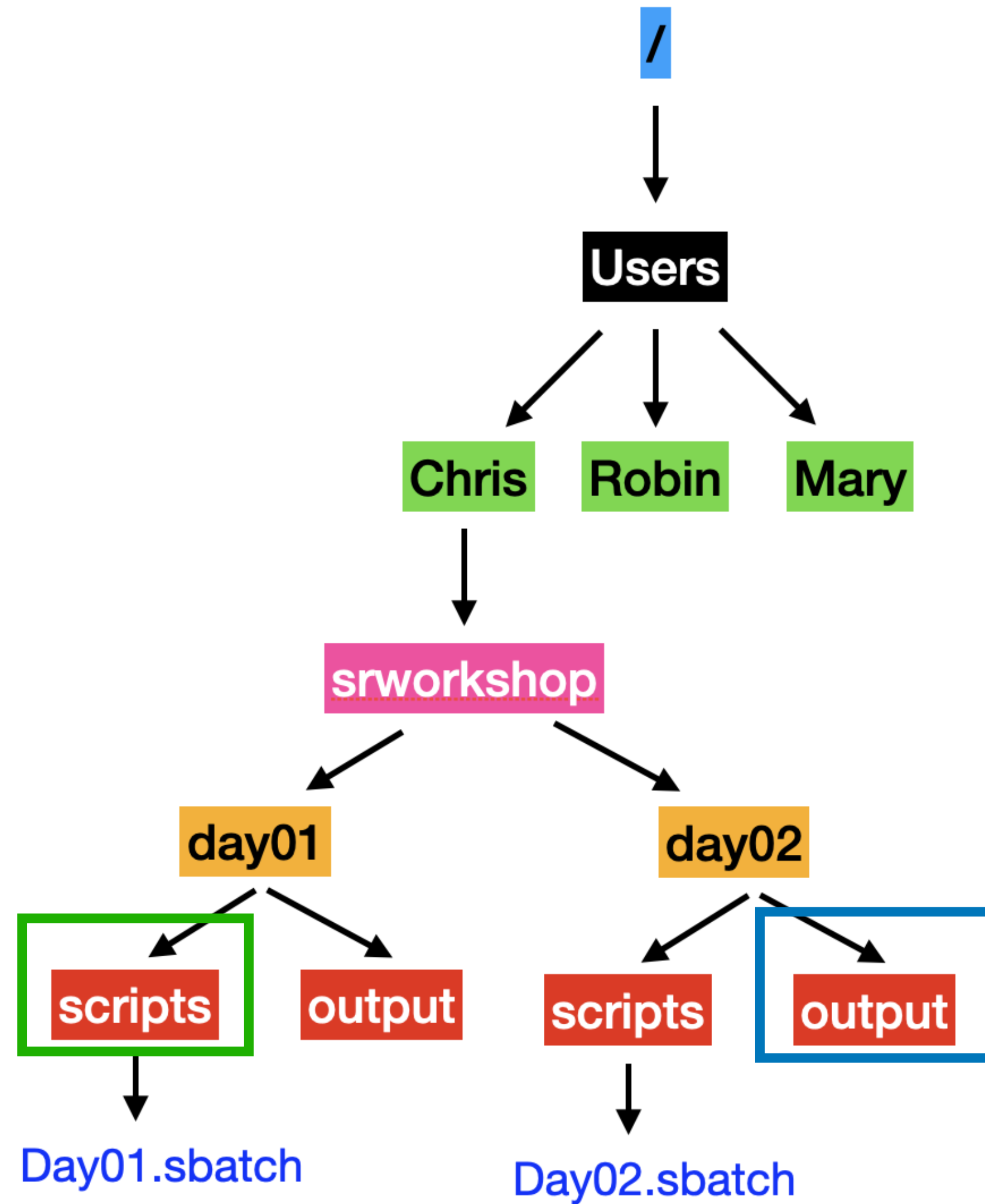
`cd /Users/Chris/srworkshop/day02`

2. Relative Path - path from working directory

`cd ../`

Move to parent directory (one directory above)

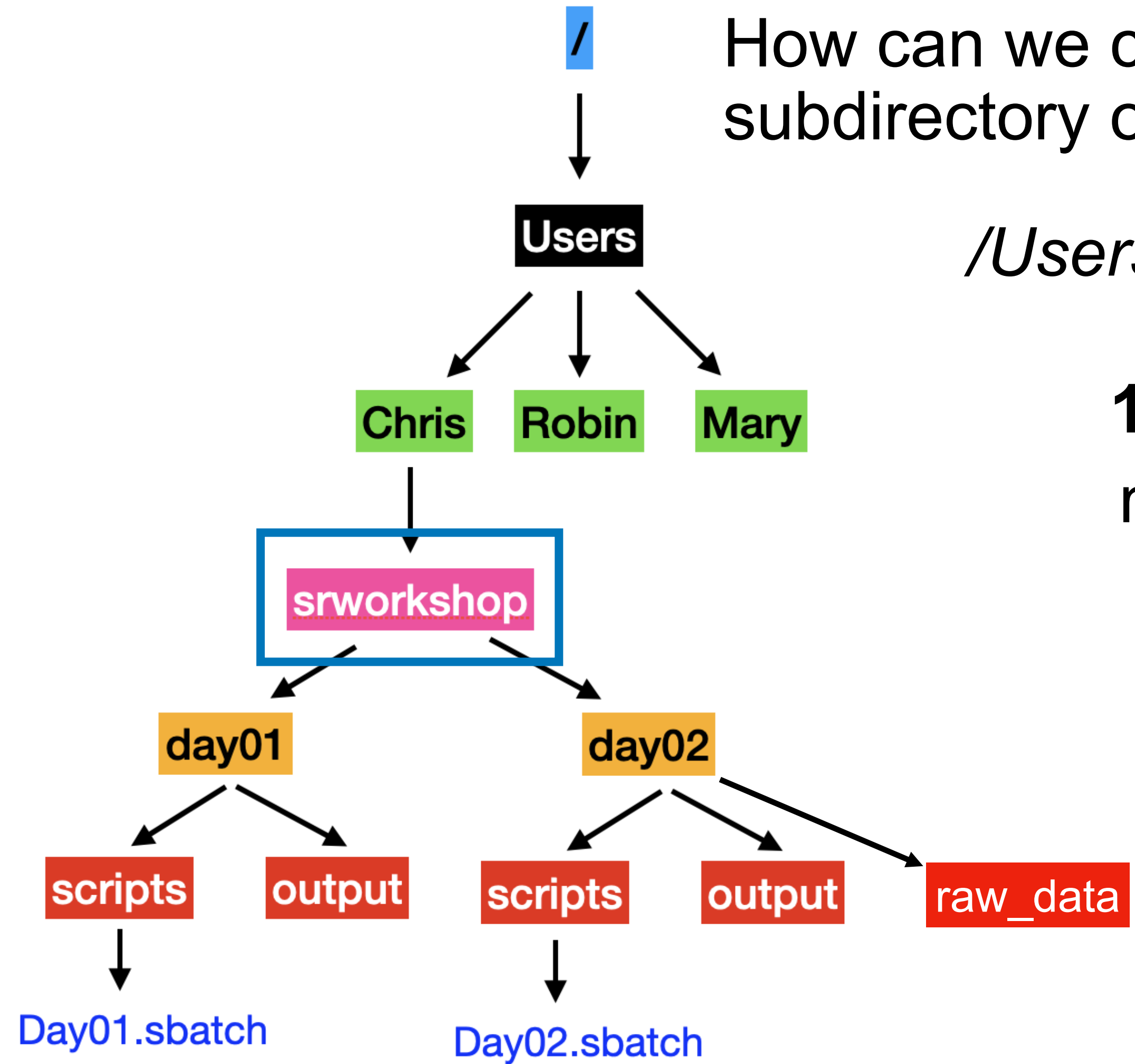
Navigating the Hierarchical Directory Structure



Write a command to navigate from day02
output to day01 scripts using a *relative path*

```
cd ../../day01/scripts
```

Creating the Hierarchical Directory Structure



How can we create a new directory called **raw_data** as a subdirectory of **day02** from the current working directory:

/Users/Chris/srworkshop

1. Absolute Path

mkdir /Users/Chris/srworkshop/day02/raw_data

2. Relative Path

mkdir day02/raw_data

3. Change Directories to Parent

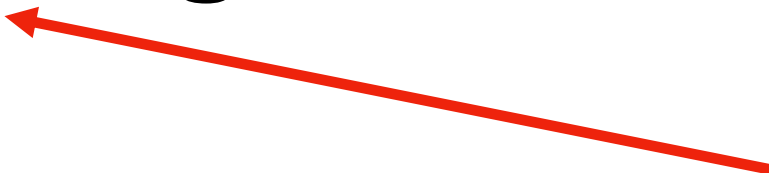
cd day02
mkdir raw_data

Bash Command Format

Options modify a command's function by adding one or more argument

How do we write a command that shows all files in the long format?

command -options arguments <input>



Flags

ls Option	Description
a	Lists all files, including hidden files beginning with a .
l	Displays results in the long format, displays permissions
r	Displays results in reverse order (Normally, ascending alphabetical)
h	In the long format, displays the human readable size of the file

1. `ls -a -l <directory>`

2. `ls -al <directory>`

3. `ls - - all -l <directory>`

Bash Command Format

To Do: Use the man command to find a complete list of the ls options for the following

1. On the AWS, navigate to /Users/<username>/srworkshop/day02
2. Print the contents of the directory to show all files in the long format
3. Now, print the contents of the directory to show the index number of each file or directory

Bash Command Format

command 1 <input> | command 2

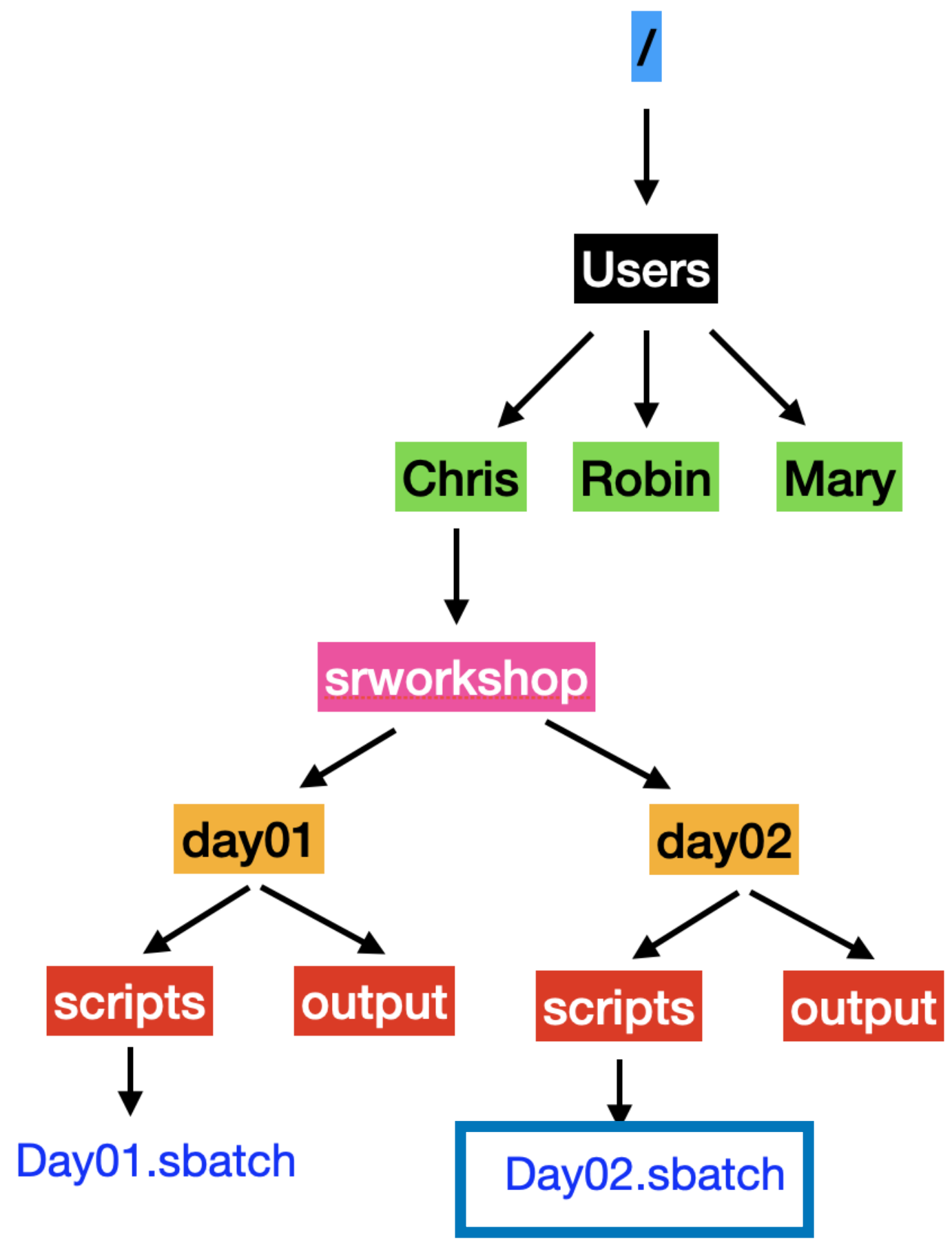
For any command in the bash command line, you can use the “|” command to have the output of the first command become the input of the second command

command <input> > <output_file>

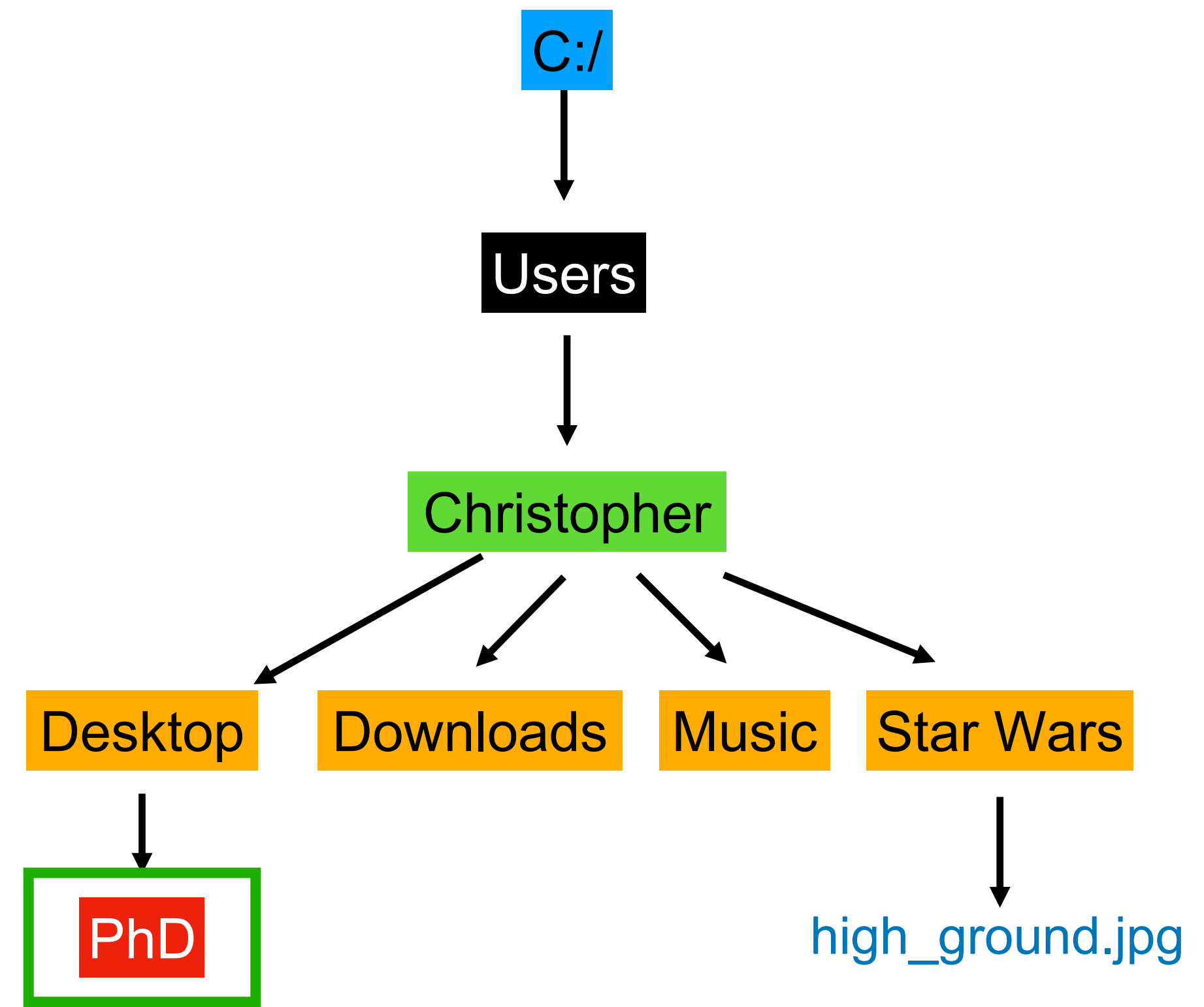
The “>” takes the output of a command series and puts it into a new output file

Copying a File from the Super Computer to the Local

AWS Super Computer



Local Computer



How to Do I copy Day02.sbatch and move it to my local PhD Directory?

Copying a File from the Super Computer to the Local

rsync <source> <destination>

On local in target directory destination

```
[christopherrauchet@Christophers-Laptop ~ % cd Desktop/PhD  
[christopherrauchet@Christophers-Laptop PhD % rsync crauchet@13.58.16.233:/Users/crauchet/srworkshop/day02/README.md ./
```

username@ip

:<absolute_path_source>

Relative path destination local

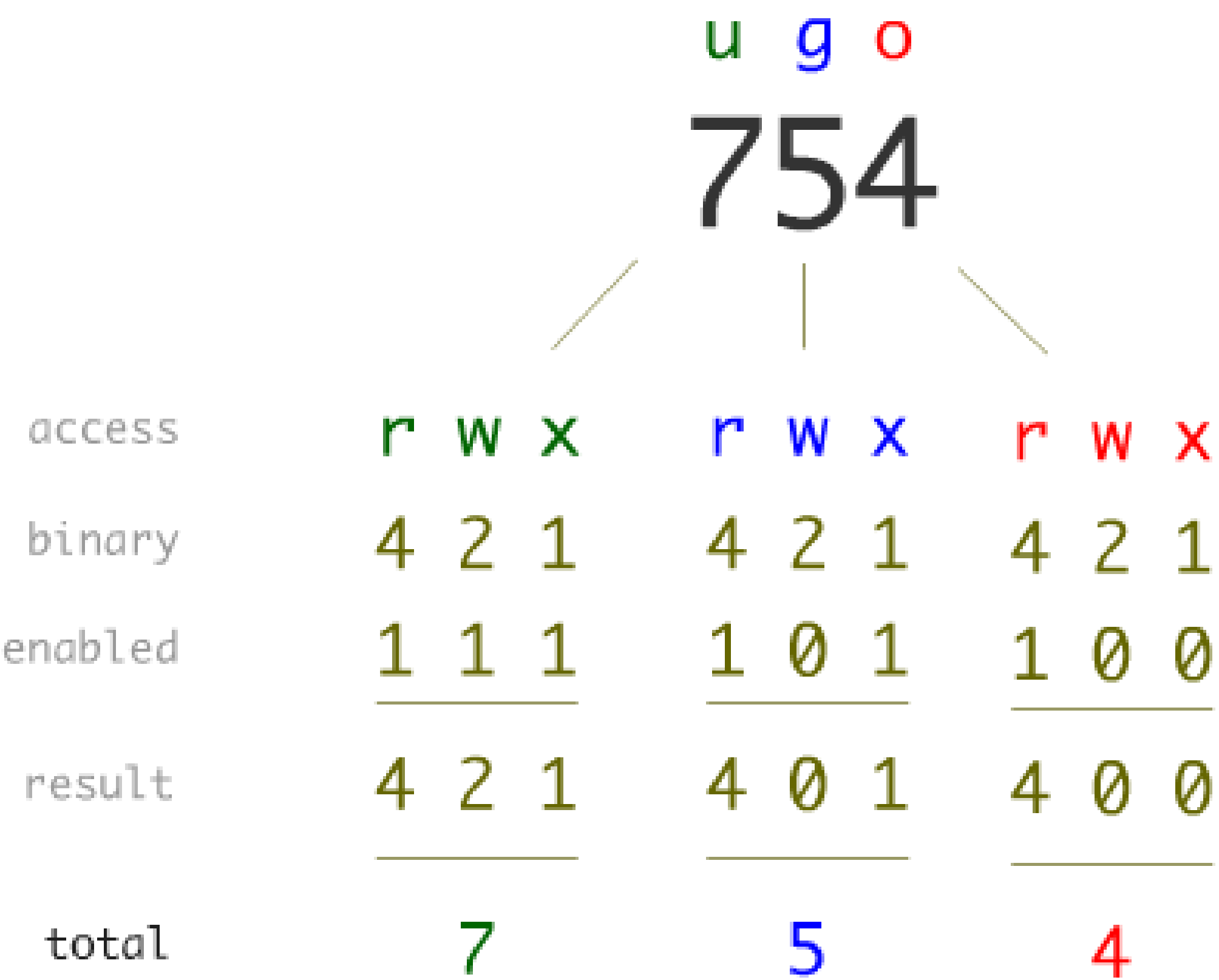
Complete Worksheet day02 worksheet.md Part 1

Note: We will be using the /scratch/ directory rather than home directory

Permissions

A file or directory's permission can be viewed using ls -l and it can be altered using chmod

chmod <permssion code> <file>



Complete Worksheet day02 worksheet.md Part 2